Name: _____

Class: _____Date:_____

LESSON 11. COMMON DIVISORS. GREATEST COMMON DIVISOR

2.33. Write the set of divisors of 66, the set of divisors of 110 and the set of common divisors of 66 and 110.

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2.34. Find the GCD of:

a) 26 and 78;

b) 18; 162 and 126.

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2.35. Find the greatest common divisor, then find the common divisors of:

a) 84 and 108;

b) 150; 225 and 105.

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2.36. Fill in the blanks with suitable words:

a) If a ⋮ 11 and b ⋮ 11, then 11 is a ..... of a and b;

b) If 13 is the largest number such that a ⋮ 13 and b ⋮ 13, then 13 is the ..... of a and b.

2.37. Tuấn and Hà each bought some boxes of colour pencils. Every box has the same number of pencils, and that number is at least 2. Tuấn bought 28 pencils and Hà bought 21 pencils. How many pencils are there in each box?

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2.38. A number is called a perfect number if it is equal to the sum of its divisors (not counting itself). For example, the divisors of 6 (not counting 6) are 1; 2; 3 and 1 + 2 + 3 = 6, so 6 is a perfect number. Which of the numbers 18; 28; 496 are perfect numbers?

By the end of 2024, only 52 perfect numbers had been found. The largest one known has more than 82 million digits.

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2.39. Find the largest natural number a such that 360 ⋮ a and 540 ⋮ a.

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2.40. Are the following fractions in simplest form? Simplify them if they are not.

a) $\frac{27}{45}$; b) $\frac{31}{67}$.

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2.41. Find all pairs of non-zero natural numbers not greater than 70 whose GCD is 23.

2.42. Find all non-zero natural numbers a and b such that a + b = 90 and GCD(a, b) = 15.

2.43. Find all pairs of non-zero natural numbers whose GCD is 6 and whose product is 540.